G - Mode in the Subtree

分值：625 分

Problem Statement

You are given a rooted tree with N vertices numbered 1 through N. Vertex 1 is the root, and the parent of vertex i is vertex p_i ($p_i < i$).
Each vertex is colored: vertex i is colored with color c_i ($1 \leq c_i \leq N$).
For each $v = 1, 2, \ldots, N$, solve the following problem:

给定一棵包含 N 个顶点的有根树，顶点编号为 1 到 N。顶点 1 是根，且对于 $p_i < i$，顶点 i 的父节点为顶点 p_i。 每个顶点都有颜色：顶点 i 的颜色为 c_i（$1 \leq c_i \leq N$）。 对每个 $v = 1, 2, \ldots, N$，求解以下问题：

Let f_i be the number of vertices colored with color i among the vertices in the subtree rooted at vertex v.

Find:

- the maximum value m among the values in the sequence $(f_1, f_2, \ldots, f_N)$, and
- the number k of positive integers i not greater than N such that $f_i = m$.

定义 f_i 为以顶点 v 为根的子树中，颜色为 i 的顶点数量。

求：

- 序列 $(f_1, f_2, \ldots, f_N)$ 中的最大值 m，以及
- 不超过 N 且满足 $f_i = m$ 的正整数 i 的个数 k。

Input/Output Format

The input and output for this problem follow a special format.

本题的输入和输出采用特殊格式。

Input Format

From Standard Input, you are given the integer N along with integers seed, M, F and $q_2, q_3, \ldots, q_M$, $d_1, d_2, \ldots, d_M$. Reconstruct $p_2, p_3, \ldots, p_N$ and $c_1, c_2, \ldots, c_N$ using the computation described by the following pseudocode. (Here, `2^31` means $2^{31} = 2147483648$. Note that state is a variable, and that a 64-bit integer type is required for computing state.)

从标准输入读取整数 N，以及整数 seed, M, F 和 $q_2, q_3, \ldots, q_M$（$d_1, d_2, \ldots, d_M$）。按照以下伪代码描述的计算过程重构 $p_2, p_3, \ldots, p_N$ 和 $c_1, c_2, \ldots, c_N$。（此处 `2^31` 表示 $2^{31} = 2147483648$。注意 state 是变量，计算 state 时需要使用 64 位整数类型。）

```
state = seed

for i=2 to N:
  if i <= M:
    p\[i\] = q\[i\]
  else:
    p\[i\] = (state mod (i-1)) + 1
    state = (state \* 1103515245 + 12345) mod 2^31

for i=1 to N:
  if i <= M:
    c\[i\] = d\[i\]
  else:
    c\[i\] = (state mod F) + 1
    state = (state \* 1103515245 + 12345) mod 2^31
```


```
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for i=1 to N:
  if i <= M:
    c\[i\] = d\[i\]
  else:
    c\[i\] = (state mod F) + 1
    state = (state \* 1103515245 + 12345) mod 2^31
```

Output Format

Let m_i and k_i denote m and k when $v = i$, respectively. Output this value:

设 m_i 和 k_i 分别为当 $v = i$ 时的 m 和 k。输出以下值:

$$\left(\sum_{i=1}^{N}(m_i \oplus i) \times (k_i \oplus i)\right) \bmod 998244353$$

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Here, $\oplus$ denotes bitwise XOR. Beware of overflow during computation.

此处 $\oplus$ 表示按位异或。计算过程中请注意溢出问题。

Constraints

- $2 \le N \le 2.5 \times 10^6$
- $1 \le p_i < i$
- $1 \le c_i \le N$
- $1 \le \mathrm{seed} < 2^{31}$
- $2 \le M \le \min(N, 10^5)$
- $1 \le F \le N$
- $1 \le q_i < i$
- $1 \le d_i \le N$
- All input values are integers.

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- $1 \le c_i \le N$
- $1 \le \mathrm{seed} < 2^{31}$
- $2 \le M \le \min(N, 10^5)$
- $1 \le F \le N$

- $1 \leq q_i < i$
- $1 \leq d_i \leq N$
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

N seed M F
$q_2\ q_3\ \ldots\ q_M$
$d_1\ d_2\ \ldots\ d_M$

Output

Output $\left(\sum_{i=1}^{N}(m_i \oplus i) \times (k_i \oplus i)\right) \bmod 998244353$.

输出 $\left(\sum_{i=1}^{N}(m_i \oplus i) \times (k_i \oplus i)\right) \bmod 998244353$。

Sample Input 1

```
4 454 4 2
1 2 2
1 2 2 3
```

Sample Output 1

```
29
```

- $i = 1 : m_1 = 2, k_1 = 1, (m_1 \oplus 1) \times (k_1 \oplus 1) = 0$
- $i = 2 : m_2 = 2, k_2 = 1, (m_2 \oplus 2) \times (k_2 \oplus 2) = 0$
- $i = 3 : m_3 = 1, k_3 = 1, (m_3 \oplus 3) \times (k_3 \oplus 3) = 4$

- $i=4: m_4=1, k_4=1, (m_4 \oplus 4) \times (k_4 \oplus 4)=25$
- $i=1: m_1=2, k_1=1, (m_1 \oplus 1) \times (k_1 \oplus 1)=0$
- $i=2: m_2=2, k_2=1, (m_2 \oplus 2) \times (k_2 \oplus 2)=0$
- $i=3: m_3=1, k_3=1, (m_3 \oplus 3) \times (k_3 \oplus 3)=4$
- $i=4: m_4=1, k_4=1, (m_4 \oplus 4) \times (k_4 \oplus 4)=25$

Thus, output $0+0+4+25=29$.

因此，输出 $0+0+4+25=29$。

Sample Input 2

```
6 123 2 2
1
1 2
```

Sample Output 2

```
101
```

The values of p_i, c_i in this test case are as follows:

该测试用例中 p_i, c_i 的值如下：

- $p=(1,2,1,2,3)$
- $c=(1,2,2,1,2,1)$
- $p=(1,2,1,2,3)$
- $c=(1,2,2,1,2,1)$

Sample Input 3

```
15 1 4 5
1 2 3
```

```
5 3 1 3
```

Sample Output 3

```
1199
```

The values of p_i, c_i in this test case are as follows:

该测试用例中 p_i, c_i 的值如下:

- $p = (1, 2, 3, 2, 1, 4, 7, 6, 5, 10, 1, 10, 2, 8)$
- $c = (5, 3, 1, 3, 4, 2, 2, 2, 4, 2, 2, 5, 3, 5, 3)$
- $p = (1, 2, 3, 2, 1, 4, 7, 6, 5, 10, 1, 10, 2, 8)$
- $c = (5, 3, 1, 3, 4, 2, 2, 2, 4, 2, 2, 5, 3, 5, 3)$